



NC STATE
UNIVERSITY

NRP 
Nuclear Reactor Program

NE235 Nuclear Reactor Operations Module #7

Prompt –vs- Delayed Neutrons & Delayed Neutron Fraction

**Dept. of Nuclear Engineering
North Carolina State University**

NE235 Nuclear Reactor Operations Training

Module #7

Module #7 Contents:

1. Prompt vs Delayed Neutron Generation Times
2. Reactor Design Considerations
3. Fission Product Delayed Neutron Pre-Cursors
4. Neutron Multiplication Cycle including Delayed Neutrons
5. Response to Reactivity at Beginning –vs- End of Core Life

Laboratory Session this week:

- ***Control Rod Calibration- Thurs. 10/20***

NE235 Nuclear Reactor Operations Training

Module #7

Module #6 Recap

Sub-Critical Multiplication

1. Equal changes in K_{eff} (or ρ) DO NOT result in equal changes in equilibrium neutron population (N_{eq})
2. Each time the difference between K_{eff} and 1 is cut in half, N_{eq} doubles (therefore SRM count rate doubles)
3. As $K_{\text{eff}} \rightarrow 1$, more generations are required to reach equilibrium

$$M = \frac{CR_2}{CR_1} = \frac{1 - K_1}{1 - K_2} \text{ Eq. \#1.26}$$

$$M = \frac{\# \text{ Neutrons from Source} + \# \text{ Neutrons From Fission}}{\# \text{ Neutrons from Source}}$$

$$\therefore \frac{1}{M} = \frac{CR_0}{CR_i} = \frac{S}{S + N_f}$$

$$\text{As } CR_i \rightarrow \infty \text{ (or } N_f \rightarrow \infty), K_{\text{eff}} \rightarrow 1 \text{ \& } \frac{1}{M} \rightarrow 0$$

Prompt vs Delayed Neutron Generation Times

Prompt Neutron Generation Time:

$$\ell_p = 3 \times 10^{-5} \text{ sec} = \text{birth time} + \text{slowing down} + \text{diffusion time}.$$

Delayed Neutron Generation Time :

$$\ell_d = 12.7 \text{ sec}$$
$$= \text{Mean life of delayed neutron precursors} + \text{Slowing down} + \text{diffusion time}.$$

Where: 99.3% Fission neutrons are from prompt; 0.7% are delayed,

$$\therefore \beta = 0.007 \frac{\Delta K}{K}$$

Neutron Lifecycle and Generation Time

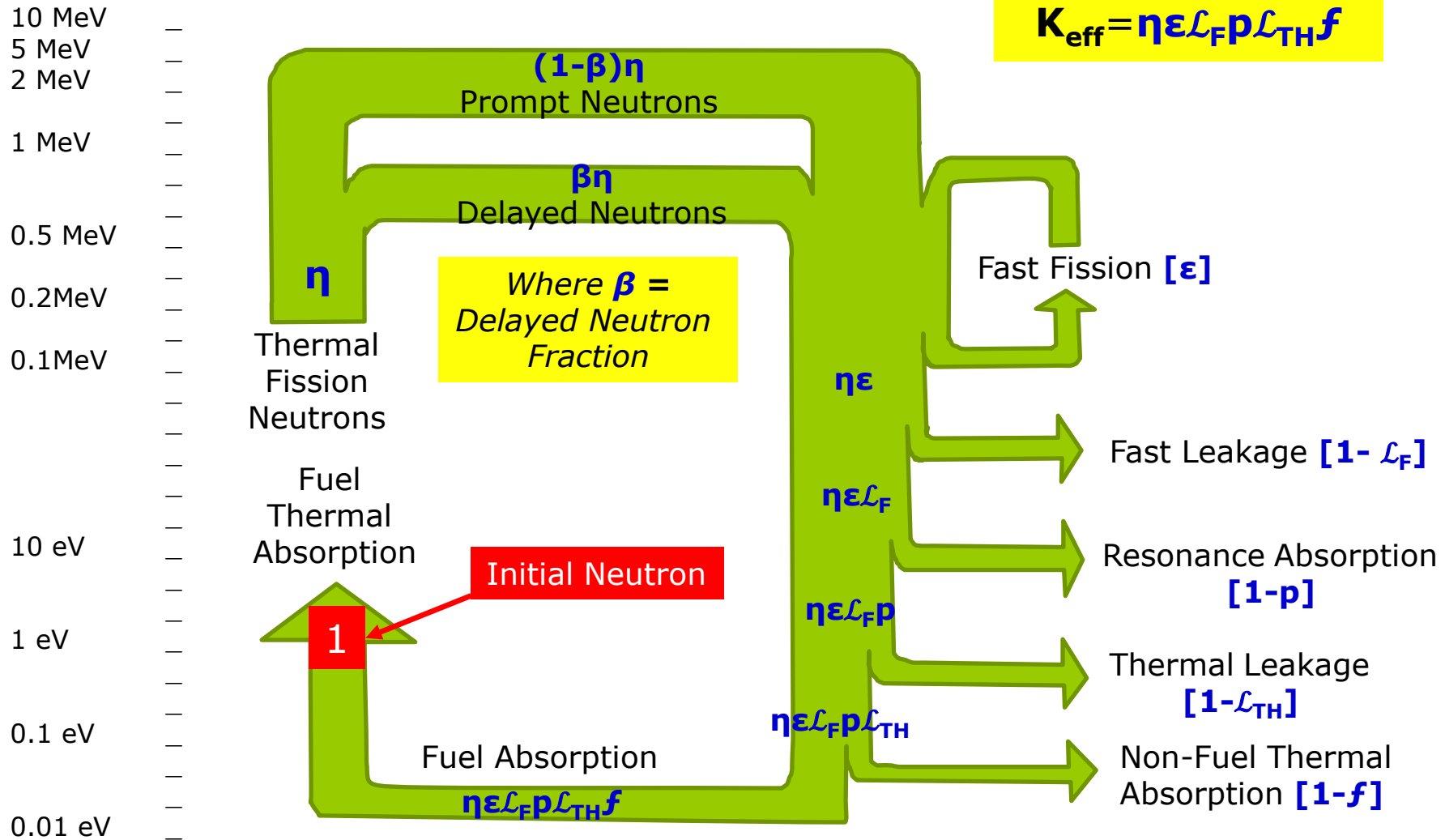
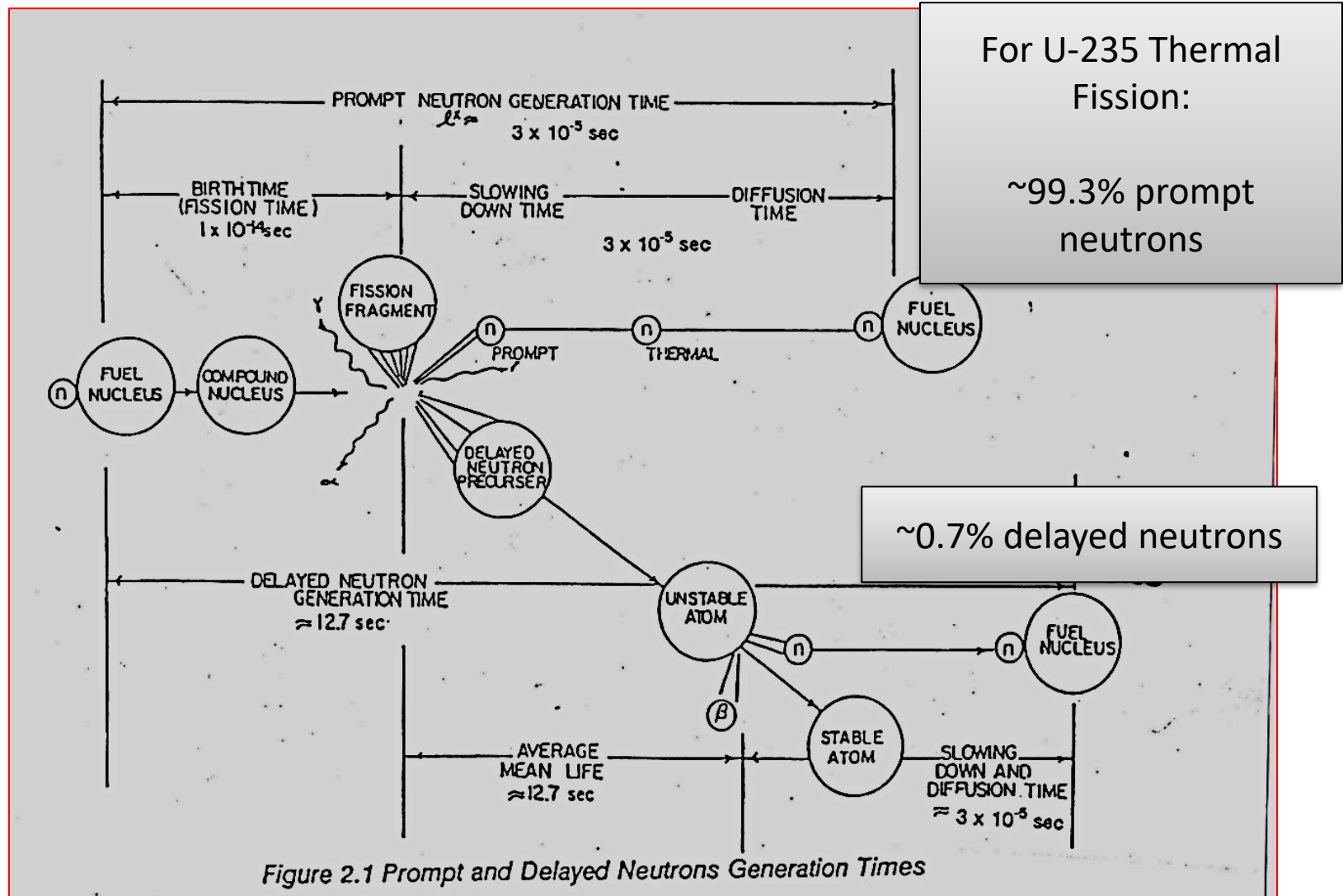


Fig. 2.1: Prompt -vs- Delayed Neutron Generation Times



Prompt vs Delayed Neutron Generation Times

So, Is the Reactor Controllable Using Prompt Neutrons Alone?

If
$$P_f = P_i e^{\left(\frac{K_{eff}-1}{\bar{\ell}}\right)t} \quad (Eq. \#2.1)$$

Where $\bar{\ell} = \text{Mean Generation Time(sec)} = \underline{3 \times 10^{-5} \text{ sec for prompt neutrons.}}$
 $P_i = 10W \quad t = 1 \text{ sec} \quad K_{eff} = 1.001001 \rightarrow \rho_{net} = 100 \text{ pcm}$

Therefore: $\bar{\ell} = \ell_p ; P_f = (10 \text{ watts}) e^{\left(\frac{1.001001-1}{3 \times 10^{-5}}\right)(1 \text{ sec})} = 3 \times 10^{15} \text{ watts}$

Assume this power is sustained for **0.1sec** => **3×10^{14}** watt-seconds of energy would be released, where 1 kiloton of TNT = 4.14×10^{12} watt-seconds.

Therefore, the total energy released would be **72 kilotons of TNT**,

where 1 Hiroshima Bomb = ~15 kilotons of TNT! ... clearly not controllable!

Prompt vs Delayed Neutron Generation Times

Taking into Account Delayed Neutrons:

If
$$P_f = P_i e^{\left(\frac{K_{eff}-1}{\bar{\ell}}\right)t} \quad Eq. \#2.1$$

Where: $P_i = 10W$; $t = 1 \text{ sec}$; $K_{eff} = 1.001001$; $\rightarrow \rho_{net} = 100 \text{ pcm}$
 and $\beta = 0.007$

If
$$\bar{\ell} = \ell_d(\beta) + \ell_p(1 - \beta) \quad Eq. \#2.3$$

Then
$$\bar{\ell} = 12.7(0.007) + 3 \times 10^{-5}(1 - 0.007) = 0.0889 \text{ sec}$$

$$\therefore P_f = (10 \text{ watts}) e^{\left(\frac{1.001001-1}{0.0889}\right)(1 \text{ sec})} = 10.11 \text{ Watts}$$

Therefore the reactor is clearly uncontrollable on prompt neutrons alone, so we design reactor to need delayed neutrons to maintain criticality!

Reactor Design Considerations

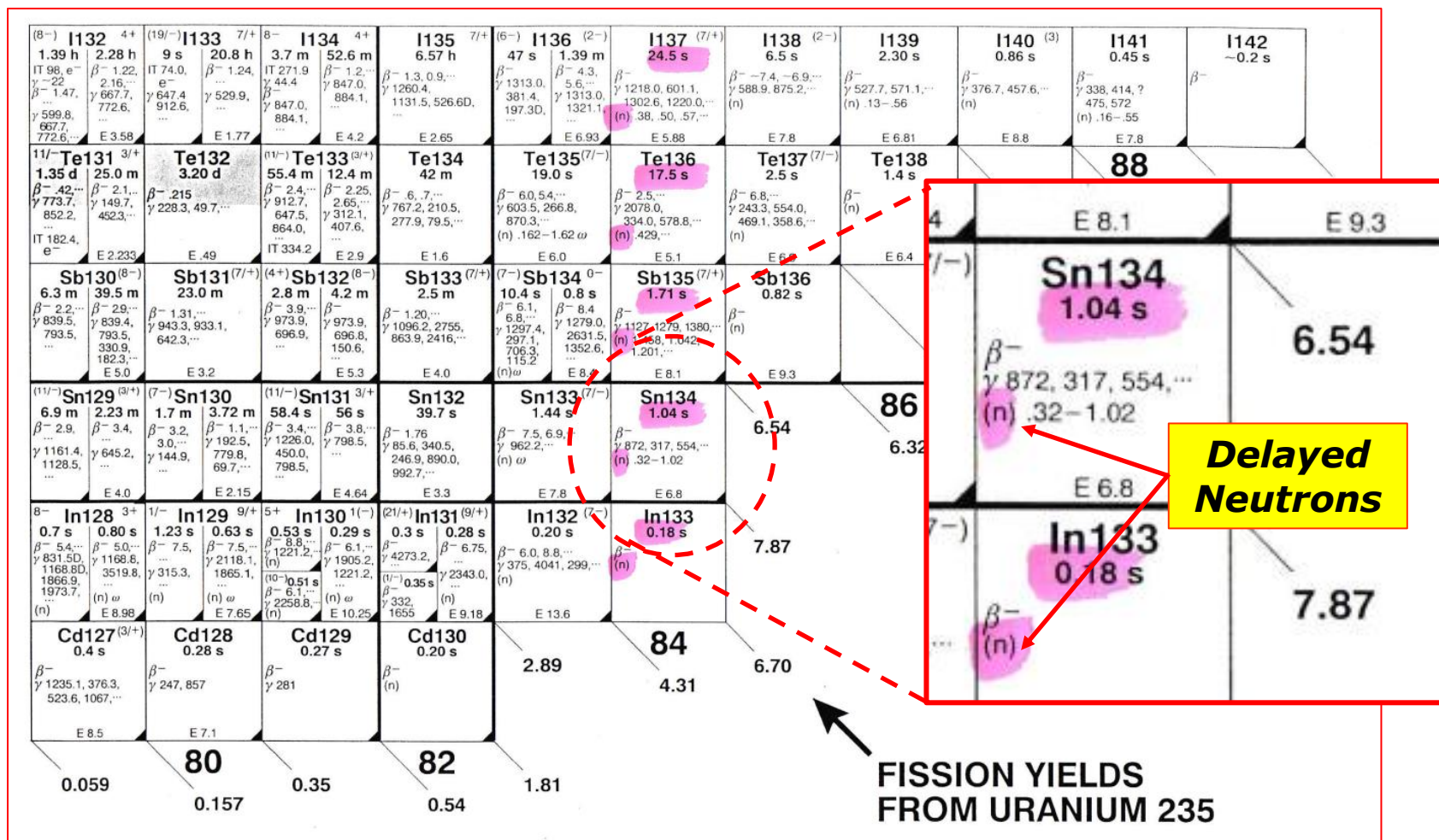
We therefore design the reactor to be sub-critical without delayed neutrons.

Adjusting parameters such as:

- **Fuel Loading** – How much ^{235}U in core + enrichment
- **Fuel to Moderator Ratio** – e.g. ^{235}U vs H_2 in core
- **Core Surface to Volume Ratio** - Adjust leakage
- **Control Rod Loading** – Absorber loading & withdrawal speed

We want “sources” of neutrons to be only slightly greater than “sinks” of neutrons so that you are subcritical without delayed neutrons.

Fission Product Delayed Neutron Pre-Cursors



Delayed Neutron Pre-Cursors – 6 Groups

Table 3.5 shows Delayed neutron data for thermal fission in ^{235}U grouped by average half-life.

For Group 1:

- ❑ The mean half-life = 55.72 sec
- ❑ The mean life $\bar{t} = \frac{55.72}{\ln(2)} = 80.4\text{sec}$ (average time for decay of precursors)
- ❑ The sub-fraction $\beta_1 = 0.000215$

For Group 6:

- ❑ The mean half-life = 0.23 sec
- ❑ The mean life $\bar{t} = \frac{0.23}{\ln(2)} = 0.33\text{sec}$ (average time for decay of precursors)
- ❑ The sub-fraction $\beta_6 = 0.000273$

Each group has a different average energy, e.g. 250→450 keV, and a different average half life leading to different times to saturation!

Delayed Neutron Pre-Cursors – 6 Groups

TABLE 3.5*

 Delayed neutron data for thermal fission in ^{235}U

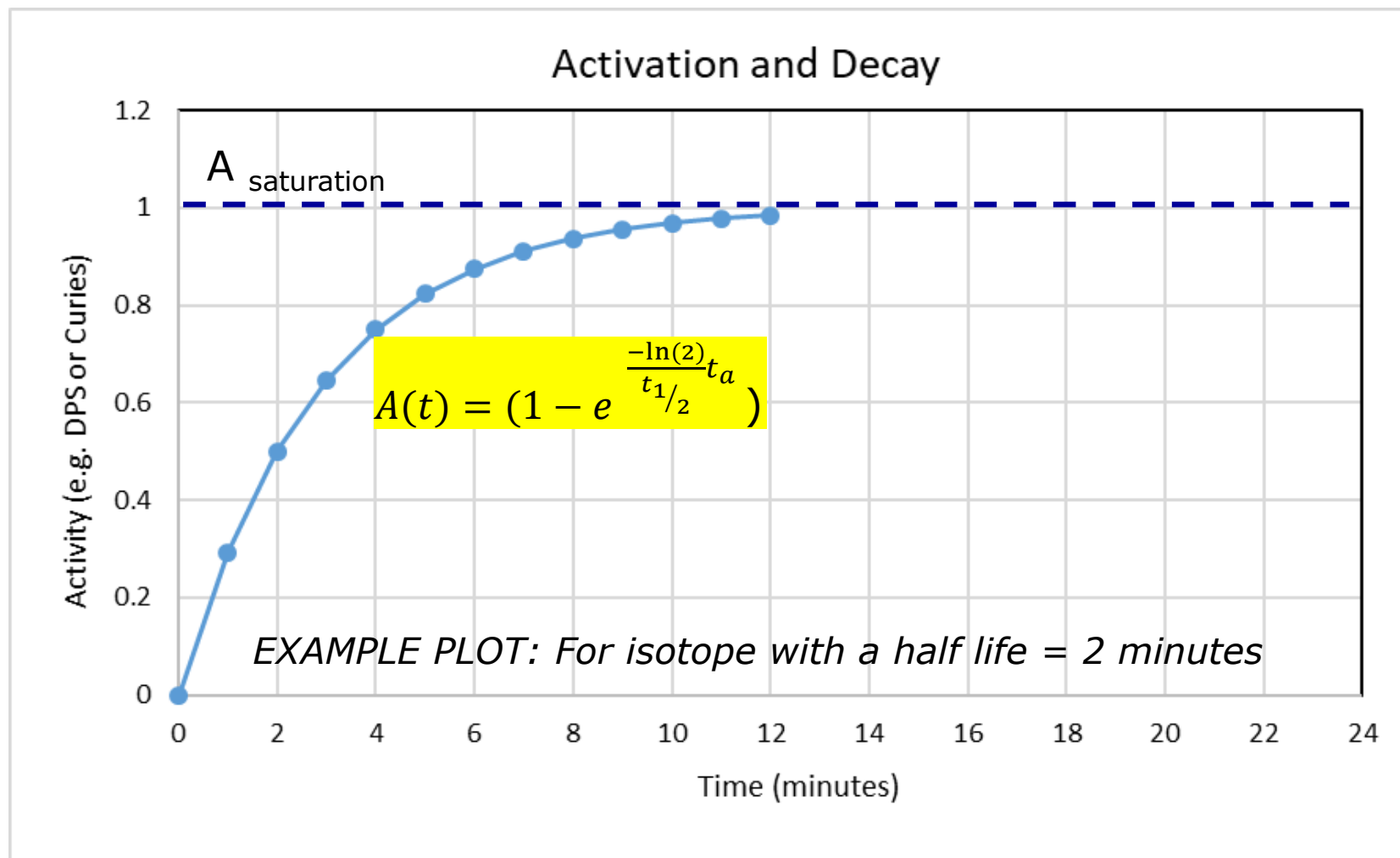
Group	Half-life sec	Decay constant $\lambda_i, \text{sec}^{-1}$	Energy keV	Yield, neutrons per fission	Fraction β_i
1	55.72 (80.4)	0.0124	250	0.00052	0.000215
2	22.72 (32.8)	0.0305	560	0.00346	0.001424
3	6.22 (8.97)	0.111	405	0.00310	0.001274
4	2.30 (3.32)	0.301	450	0.00624	0.002568
5	0.610 (0.380)	1.14	—	0.00182	0.000748
6	0.230 (0.332)	3.01	—	0.00066	0.000273

WEIGHTED AVERAGE = 9.03 (13.0) - SEC
 ↑ ↓
 MEAN LIFE MEAN HALF LIFE

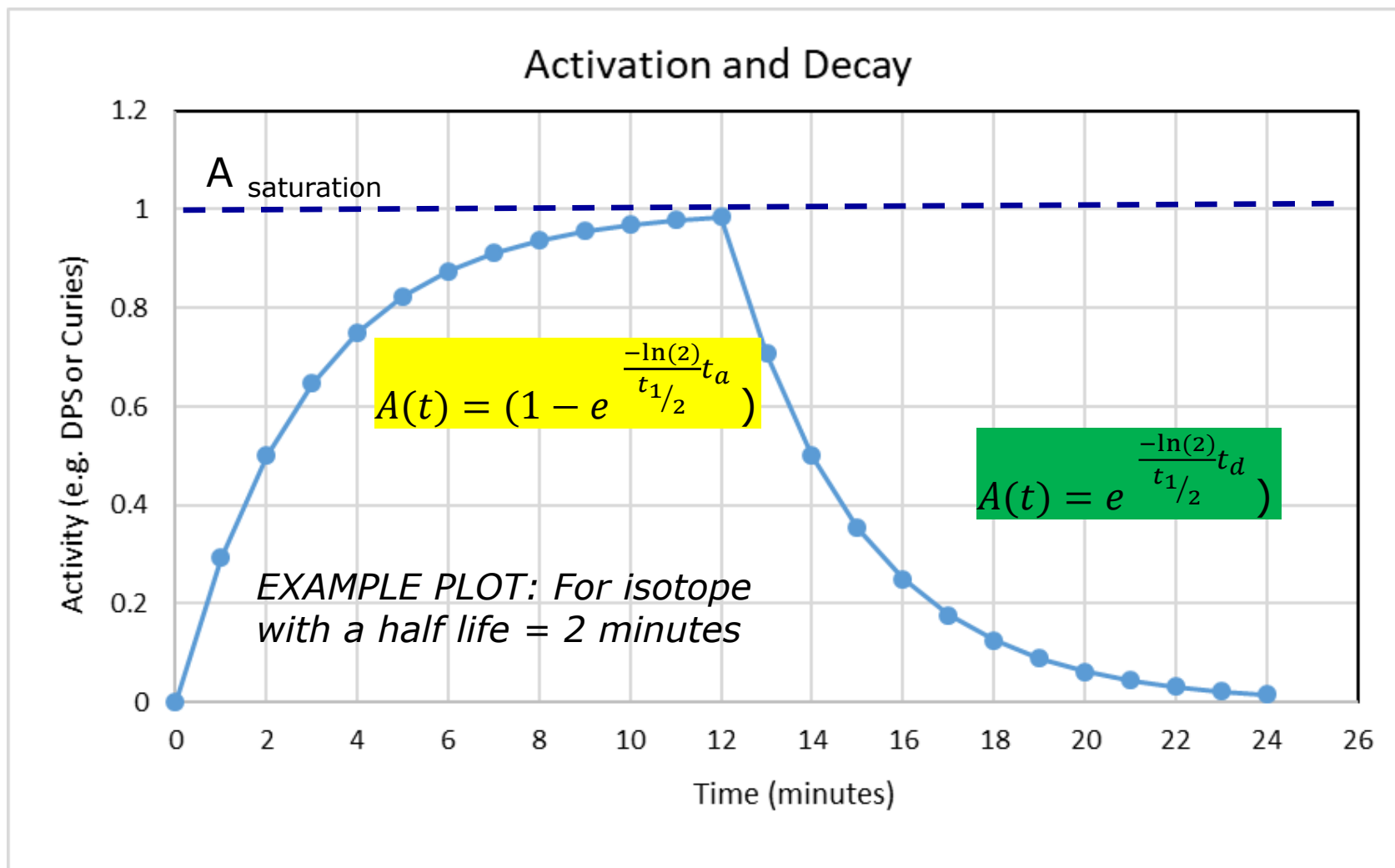
Total yield: 0.0158
 Total delayed fraction (β): 0.0065

*Based in part on G. R. Keepin, *Physics of Nuclear Kinetics*, Reading, Mass.: Addison-Wesley, 1965.

Time to Saturation: Activity –vs- Time

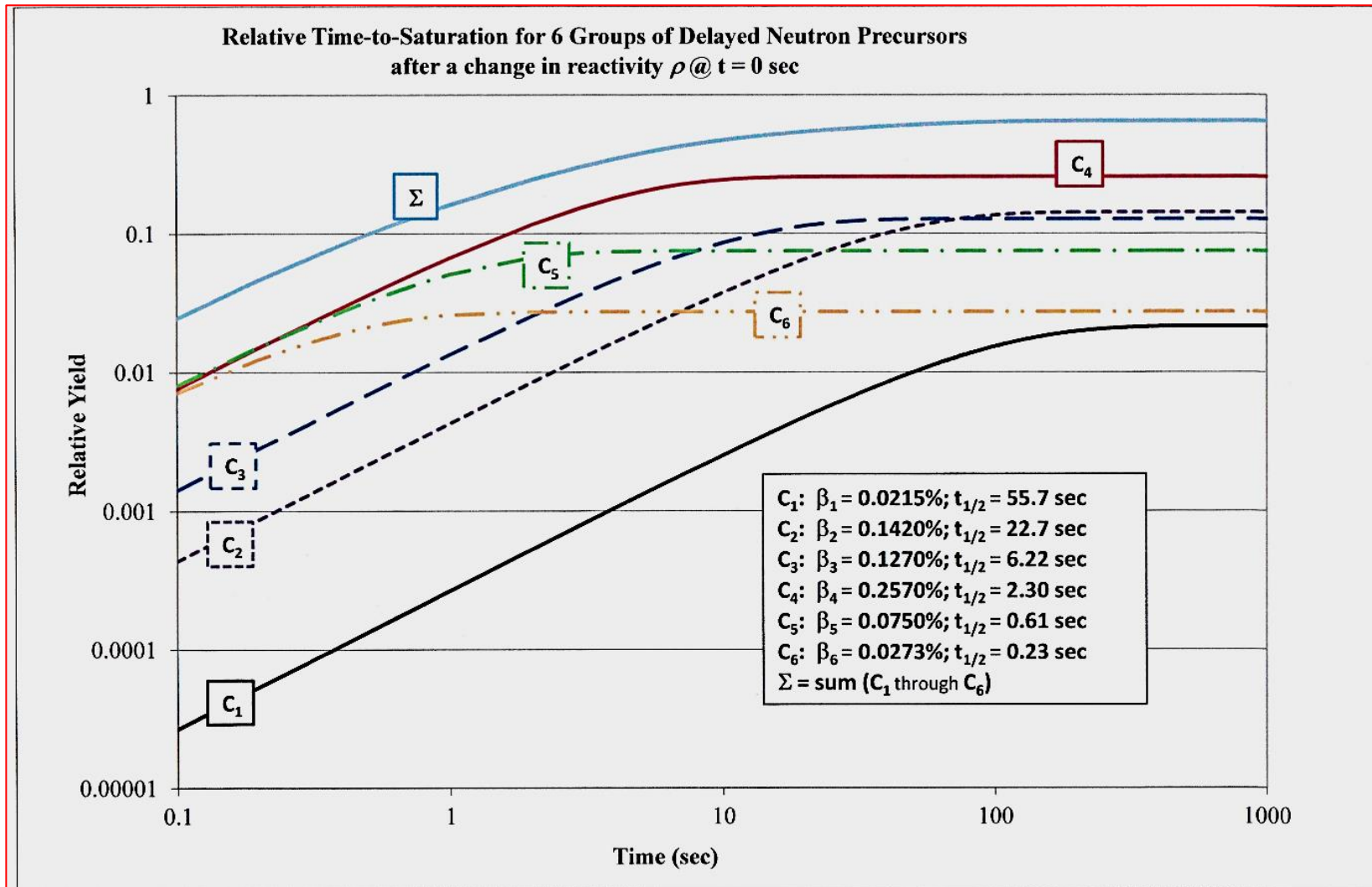


Time to Saturation: Activity –vs- Time



Delayed Neutron Precursor Groups

Time to Saturation – *based on mean half life*



Neutron Multiplication Cycle

Now let's look at how groups of precursors affect the neutron multiplication cycle for ^{235}U delayed neutrons (*reference Figure 2.2 on next slide*).

At any given instant in core we have:

A. Withholding of a portion of fission neutrons as delayed pre-cursors:

$\beta = \sum_{i=1}^6 \beta_i = 0.7\%$ stored away. This leads to C_i in storage.

B. Release of a portion of delayed neutrons from storage due to decay of precursors:

Release rate = $\sum_{i=1}^6 \lambda_i C_i$, where $C_i \propto \beta_i N \phi \sigma_f (1 - e^{-\lambda_i t_a})$

When $K_{\text{eff}} = 1$, then **neutron population is constant** and the withholding of fission neutrons as delayed precursors (A) is equal to the release of delayed neutrons from storage (B); **i.e. A = B**

Fig. 2.2 - Neutron Multiplication Cycle

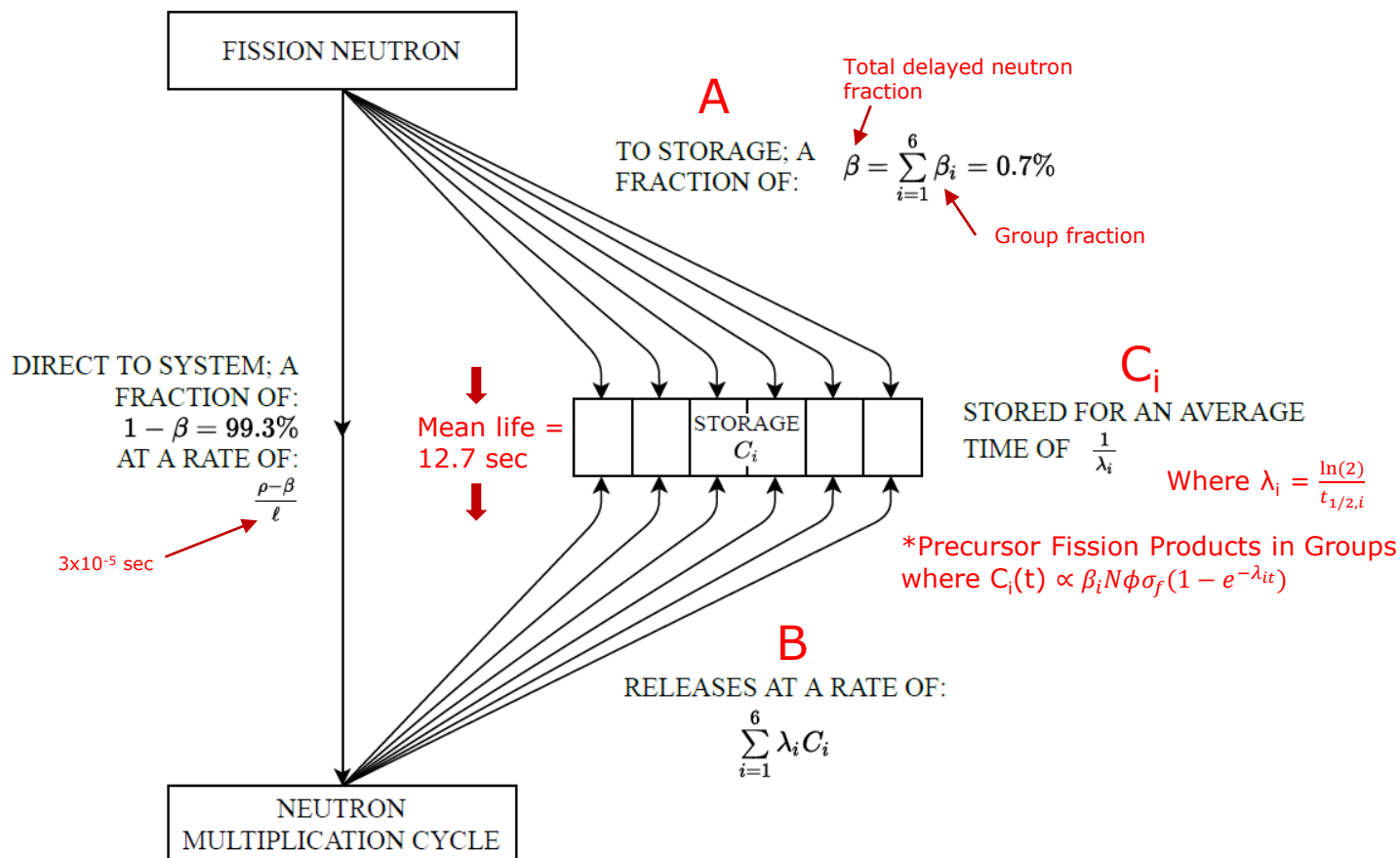


Figure 2.2: Neutron multiplication cycle.

Neutron Multiplication Cycle

If K_{eff} is raised to be > 1 : (See power trace in Fig 2.3 on next slide)

1. Initially there is a sharp rise in neutron population due to rapid multiplication of prompt neutrons.
2. The initial rate of delayed neutron release (**B**) stays the same because saturated activities (C_i) of precursor groups is based on pre-increase withholding rate (**A**). This is based on flux.
 - Withholding (**A**) builds inventory over time which leads to an increased release (**B**).
 - Therefore, (**A**) may increase immediately with prompt jump, but inventory takes time to accumulate and to result in increased release rate (**B**).
 - Therefore, the rate of population multiplication decreases as reactor 'waits' for delayed neutron release rate (**B**) to catch up.
3. Rate of release (**B**) increases as precursor group activity (C_i) increases.

For $K > 1.0$, the rate of population increase becomes exponential.

Fig. 2.3 – Reactor Response to Positive Reactivity

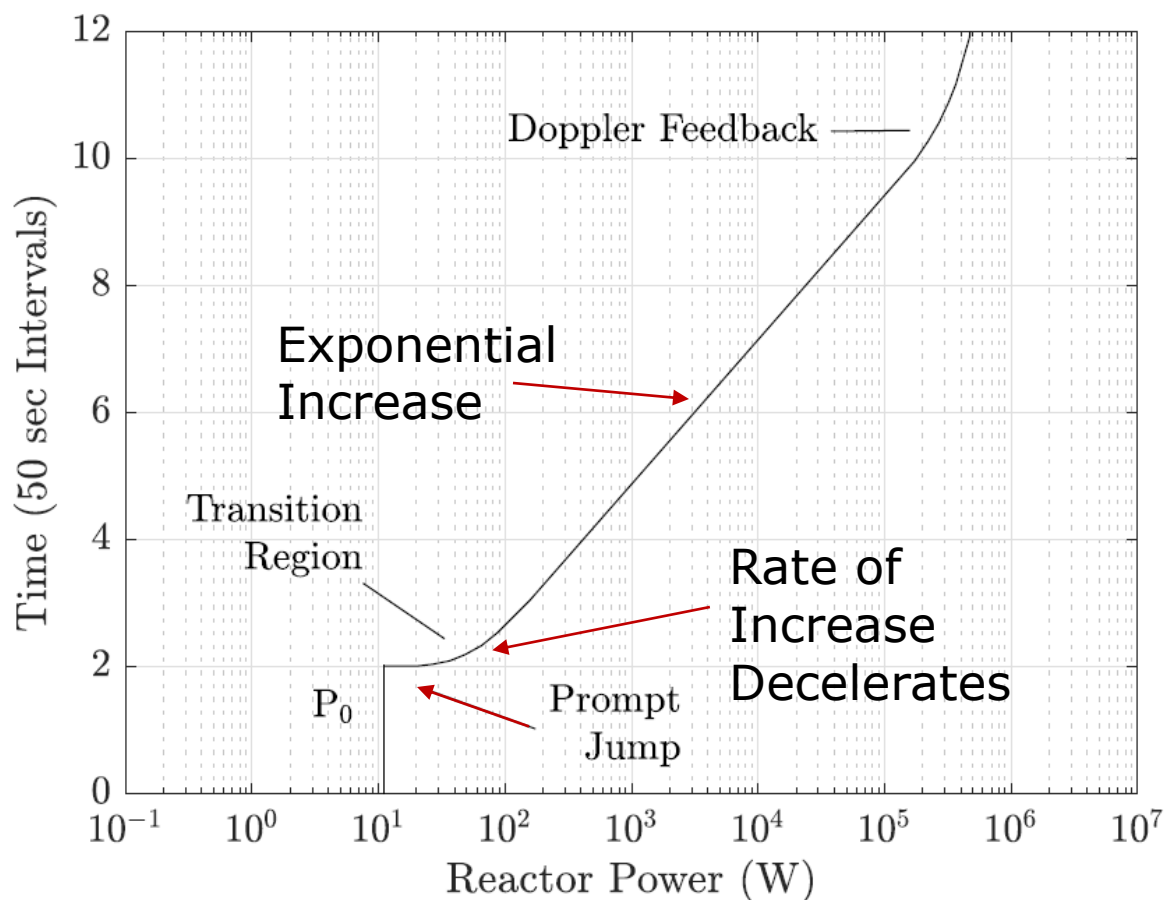


Figure 2.3: Reactor response to positive reactivity.

β at Beginning –vs- End of Core Life

So, we've seen the importance of having delayed neutrons for being able to control the rate of increase of neutron population.

The ratio of the number of Delayed neutrons vs the number of Prompt neutrons is expressed by the **Delayed Neutron Fraction: β**

$$\text{Where } \beta = \frac{\# \text{ } n' \text{ s Delayed}}{\# \text{ } n' \text{ s Prompt} + \# \text{ } n' \text{ s Delayed}}$$

β is dependent on the isotopic mix of fissile isotopes:

$$\beta^{235}\text{U} = 0.007 ; \beta^{239}\text{Pu} = 0.0025$$

This changes over core life:

$$\beta_{\text{eff}} @ \text{BOL} \approx 730\text{pcm}$$

$$\beta_{\text{eff}} @ \text{EOL} \approx 548\text{pcm}$$

(Nominal for 4% enriched PULSTAR core)

Reactor Response at Beginning –vs- End of Core Life

$$\text{SUR (DPM)} = 26.1 / \tau_{\text{sec}}$$

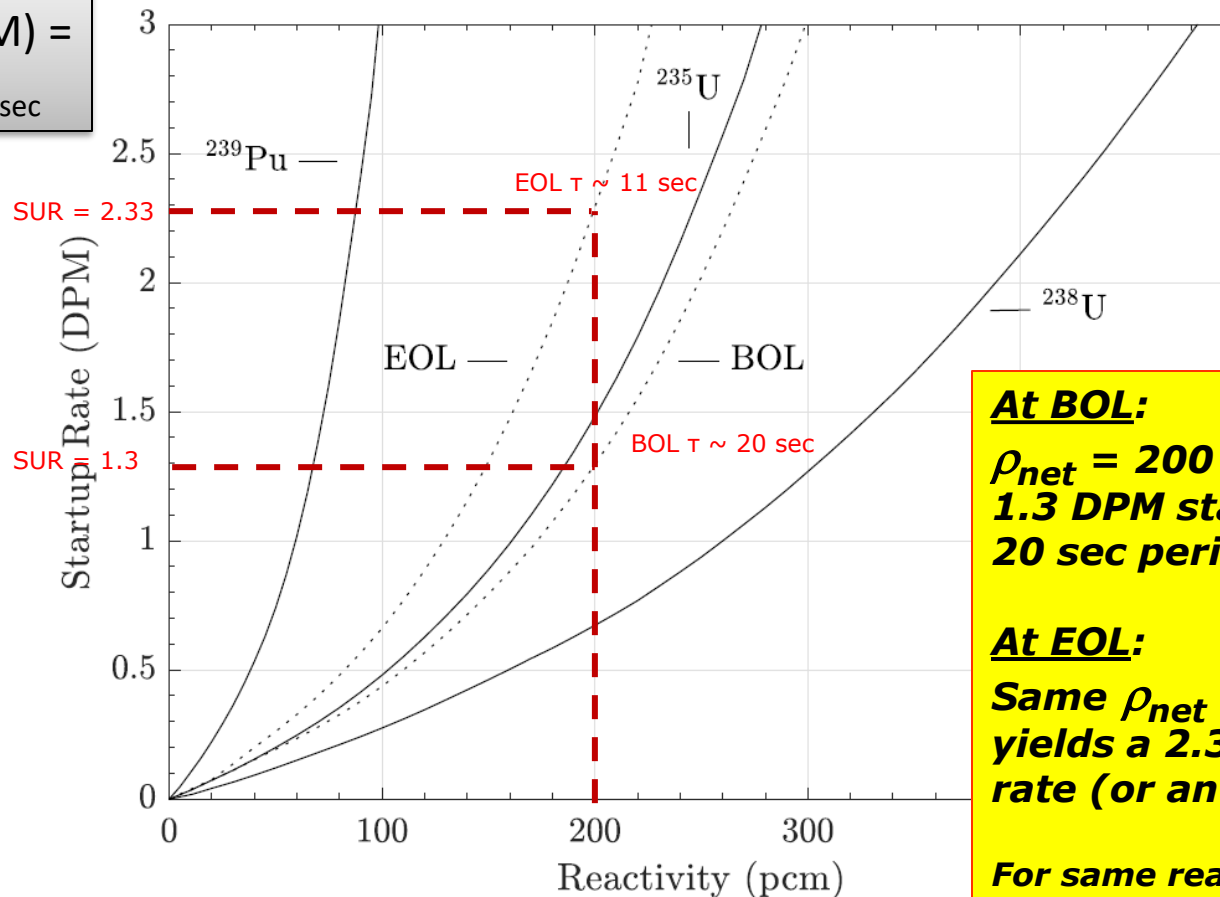


Figure 2.4: Reactor fuel characteristics.

At BOL:

$\rho_{\text{net}} = 200$ pcm yields a 1.3 DPM startup rate (or a 20 sec period).

At EOL:

Same $\rho_{\text{net}} = 200$ pcm yields a 2.3 DPM startup rate (or an 11 sec period).

For same reactivity, rate of power change increases over core life as ^{239}Pu builds in and ^{235}U depletes!

Summary

- Prompt vs Delayed Neutron Generation Times: *Delayed neutrons are required for control of the reactor.*
- Reactor Design Considerations: *Adjust design parameters to maintain reactor sub-critical without delayed neutrons.*
- Fission Product Delayed Neutron Pre-Cursors: *Delayed neutrons come from decay of neutron rich fission products.*
- Neutron Multiplication Cycle including Delayed Neutrons: *Delayed neutrons are stored and released at rates that are a function of their half-lives and fission yields.*
- Response to Reactivity at Beginning –vs- End of Core Life: *Reactor response to positive reactivity speeds up as core ages due to breeding of ^{239}Pu and depletion of ^{235}U .*